

Course Home (/learn/predictive-analytics/home/welcome) > Week 4 (/learn/predictive-analytics/home/week/4) > Kaggle Competition Peer ...

Assignment: Kaggle Competition Peer Review

Submit by November 22, 11:59 PM PT

Ready to start your next assignment?

You'll find instructions and a place to submit below.

Instructions (/learn/predictive-analytics/peer/7A5tl/kaggle-competition-peer-review)

My submission (/learn/predictive-analytics/peer/7A5tl/kaggle-competition-peer-review/submit)

Discussions (/learn/predictive-analytics/peer/7A5tl/kaggle-competition-peer-review/discussions)

Instructions

Experience an active Kaggle (<https://www.kaggle.com>) competition -- where some of the most interesting business problems are presented for data scientists to work on!

Overview of the assignment

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Kaggle (<https://www.kaggle.com/about>) is the world's largest community of data scientists. They compete with each other to solve complex data science problems, and the top competitors are invited to work on the most interesting and sensitive business problems from some of the world's biggest companies through competitions. In this assignment, you will be participating in an active Kaggle (<http://kaggle.com/>) competition. (Note: Sign up for a Kaggle account if you do not already have one.)

Goal: The purpose of this project is not that you "solve" a given Kaggle competition - but that you experience approaching a problem, creating a solution, evaluating it, and making some attempt on improving it.

Getting Started:

1. **Browse the active competitions** (<https://www.kaggle.com/competitions>) by using the "Competitions" button at the top (be sure to sign in).
2. **Select a project** that interests you and you have an idea of how to approach. If you are unsure what to choose, try the Titanic Competition (<http://www.kaggle.com/c/titanic-gettingStarted>). There is some great help available for that competition in the kaggle forums, and we discuss it in the lectures as well.

Note: This project is VERY open-ended. You may use whatever combination of techniques and tools you think are appropriate to generate a solution.

What to do next:

You will turn in a brief report of your work on the kaggle competition in 5 pieces as outlined below. Note that you will report on both an initial solution, then, after analyzing it, an improved solution.

- **Part 1: Problem Description.** Give the name of the competition you selected and write a few sentences describing the competition problem as you interpreted it. You want your writeup to be self-contained so your peer-reviewer does not need to go to Kaggle to study the competition description. Clarity is more important than detail. What's the overall goal? What does the data look like? How will the results be evaluated?

Example: "The task is to predict whether a given passenger survived the sinking of the Titanic based on various attributes including age, location of the passenger's cabin on the ship, family members, the fare they paid, and other information. Solutions are evaluated by comparing the percentage of correct answers on a test dataset." your problem, your solution, your evaluation, and some improvement. These reports will be graded by peer review.

- **Part 2: Analysis Approach.** Write a few sentences describing how you approached the problem. What techniques did you use? Clarity is more important than technical depth in this exercise.

HINT: Favor simple approaches over complex approaches. The idea is to get a good, simple, comprehensible solution rather than try to impress people with how smart you are. Start small, and improve your solution incrementally -- don't try to save the world, boil the ocean, or any other hyperbolic metaphors.

Example: "I split the data by gender and handled each class separately. For the females, I trivially classified all of them as "survived." For the males, I trained a random forest as a classifier. I ignored the pclass attribute that indicated the location of the passenger's cabin because I didn't think it was relevant."

- **Part 3: Initial Solution.** Write a few sentences describing how you implemented your approach. Think of it as a whiteboard conversation or a descriptive forum post rather than a full technical report. Try to provide enough detail for someone with some experience to follow your recipe

and reproduce your results: Describe how you prepared the data, the method(s) you applied, and any tools you used (not detailed code). What languages and libraries did you use? What challenges did you run into?

Example: "I partitioned the data by gender manually using Excel. I used Weka to build the random forest."

- **Part 4: Initial Solution Analysis.** Write a few sentences assessing your approach. Did it work? What do you think the problems were?

Example: "My approach did not work so well, achieving a score of 0.65. This is less than the sample solution. I suspect I should not have ignored the pclass attribute."

- **Part 5: Revised Solution and Analysis.** Write a few sentences describing how you improved on your solution, and whether or not it worked.

Example: "I included the pclass attribute and ignored the ticket number attribute. My score improved to 0.68."

If you've already completed kaggle competitions, you may briefly write up your solution and you're done! But of course, it may be fun to try a new one...

Have fun with this one; participating in these competitions can be very rewarding!

◀ (/learn/predictive-analytics/lecture/fHyWF/dbscan-variable-density-and-parallel-algorithms)

▶ (/learn/predictive-analytics/outline)

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